

Jaxson Gauthier

815-764-2685 | jaxsongauthier702@gmail.com | jaxsongauthier.com

Education

The University of Iowa | Iowa City, IA

Expected May 2027

Bachelor of Science and Engineering – Computer Science and Engineering - Focus area: Bioimaging

Cumulative GPA: 3.62/4.00

Relevant Coursework: Algorithms, Data Structures, Machine Learning, Deep Learning in Medical Imaging, Intro to Digital Design

Certifications: Ai for Medicine Specialization, Supervised Machine Learning: Regression and Classification (Deeplearning.ai)

Skills

Technical: Python, C/C++, NumPy, Pandas, Git, HTML/CSS, MATLAB, Pytorch, TensorFlow, SQL, JavaScript, Django, AWS, spaCy, OpenCV

Work Experience

Full-Stack Software Engineer Intern

May 2025 – July 2025

TrickyLeaf | Milan, Italy

- Built scheduling and reviewing system for snowboarding mentors using **Node.js** and role-based access, with automated **SQL-backend** lesson booking and feedback.
- Developed IoT system enabling remote control of lights using web-interface, and real-time gate lock detection using **OpenCV** integrated with a **RaspberryPi** physical backend and **AWS IoT**.
- Appointed by supervisor to mentor a fellow intern; provided foundational training and ongoing support to ensure successful project execution.

Undergraduate RA – Bioimaging & Deep Learning

November 2023 – Present

University Of Iowa, with Dr. Reinhard Beichel

- Research on quantitative PET/CT **image analysis** algorithms to evaluate and predict patient outcomes in head and neck cancer using TCIA data.
- Applied gradient-based interpretability methods on a neural network achieving a c-index of 0.95 to identify image features driving predictions while streamlining the pipeline for efficient integration of new datasets.

IT Intern, Student Technology Department

July 2022 – June 2023

Lockport Township Highschool | Lockport, Illinois

- Repaired and maintained student computers for a high school with **3000** student body count.
- Created detailed **documentation** for new problems and solutions, improving team knowledge sharing and future issue resolution.
- Updated and maintained school-wide scheduling software to ensure accurate placement of students.

Leadership Experience

Engineering Ambassador

June 2024 – Present

College of Engineering | Iowa City, IA

- Hosted events tailored for over **2000** prospective high school students through a semester whilst acting as point of communication.
- Organized and assisted in the implementation of college events including main admissions event hosting over **500** people.

Supplemental Instructor

June 2024 – May 2025

University of Iowa | Iowa City, IA

- Facilitated collaborative learning sessions for students to enhance their course knowledge in “Introductory to Engineering Problem Solving”.
- Hosted over **50** sessions assisting over **100** unique students over the course of the semester.

Technical Projects

Frequent Listener

- Built web app where users guess songs from progressively reconstructed **FFT audio signals** using **Python** and **React.js**.
- Designed **PostgreSQL** schema to track user stats, song analytics, and game progress across sessions.
- **Developed** an admin interface leveraging Spotify and YouTube APIs to streamline song ingestion, automate FFT/Inverse FFT transformations, and generate composite signals based on the **100–1000** most important frequencies.

Mednote

- Developed full-stack web application using **Django** and **React** that allows users to upload doctor’s notes and view a personal health timeline, including past visits, medications, allergies, and upcoming appointments.
- Integrated **spaCy NLP pipelines** to automatically extract key medical information (e.g., diagnoses, medications, allergies) from unstructured doctor notes and update the Django database accordingly.

Custom Animal Species Learning Model

- Developed a multi-class image classification model to identify 90 animal species using a Kaggle dataset of 5400 Images using a **ResNet-18** architecture with transfer learning and optimization using **Skorch** and cross-entropy loss.
- Achieved 78.4% test accuracy and validated model using manually collected images.

Activities, Recognition, and Certifications

Electrical Engineering Undergraduate Scholar Award	Fall 2023 - Present
Theta Tau Omicron Professional Fraternity - Active Member	August 2024 – Present
Iowa Triathlon Club - Active Member	August 2025 - Present
Advantage Iowa Award	August 2023